Transfer Learning for Wildlife Image Classification

1. Introduction

The objective of this assignment is to build an image classification model using transfer learning with MobileNet to identify different wildlife species from images. Transfer learning allows the use of a pretrained model to improve performance and reduce training time, especially when working with limited datasets. This project focuses on applying preprocessing, data augmentation, model building, training strategies, and evaluation techniques to develop an efficient wildlife classifier.

2. Dataset Description

The dataset used in this project contains 10 animal classes:

- cane (dog)
- cavallo (horse)
- elefante (elephant)
- farfalla (butterfly)
- gallina (chicken)
- gatto (cat)
- mucca (cow)
- pecora (sheep)
- ragno (spider)
- scoiattolo (squirrel)

The dataset consists of 26,179 images organized into class-wise folders. Each class contains more than 100 images, satisfying the assignment requirement.

The dataset was split into:

- Training set: 70%

- Validation set: 20%
- Test set: 10%

3. Data Preprocessing & Exploration

Images were loaded using tensorflow. keras.preprocessing.image_dataset_from_directory. All images were resized to 224×224 pixels, which is the required input size for MobileNet.

The dataset was shuffled to ensure randomness and avoid bias during training. A visualization step was performed to display 5 sample images per class, helping to understand the dataset distribution and image characteristics.

4. Data Augmentation

To improve model generalization and reduce overfitting, multiple data augmentation techniques were applied to the training data:

- RandomFlip (horizontal & vertical)
- RandomRotation (0.2)
- RandomZoom (0.2)
- Random Contrast (0.2)
- Random Translation (0.1)
- RandomBrightness (0.2)
- Gaussian Noise (optional enhancement)

These transformations artificially increased dataset diversity and made the model robust to variations such as rotation, lighting, and scale changes. Augmentation was applied only to the training dataset, not validation or test data.

5. Model Building

5.1 MobileNet as Base Model

MobileNetV2 pretrained on ImageNet was used as the base model with:

- include_top=False
- weights='imagenet'

The base model layers were frozen to retain learned features and prevent overfitting.

5.2 Custom Layers

The following layers were added on top:

- **GlobalAveragePooling2D** → Reduces feature maps into a single vector
- **Dense (128, ReLU)** → Learns complex feature relationships
- **Dropout (0.3)** → Prevents overfitting
- **Output Dense (Softmax)** → Produces class probabilities

5.3 Why MobileNet?

MobileNet was chosen because it is a lightweight and efficient model designed for low computational cost. It uses depthwise separable convolutions, which significantly reduce parameters while maintaining accuracy. This makes it ideal for real-world applications such as wildlife classification.

6. Training with Early Stopping

The model was compiled using:

- Optimizer: Adam
- Loss: Sparse Categorical Crossentropy
- Metric: Accuracy

Training was performed for a maximum of 50 epochs with early stopping:

- Monitor: validation loss
- Patience: 5 epochs
- Restore best weights: True

Early stopping helped prevent overfitting by stopping training when validation performance stopped improving.

7. Results & Plots

7.1 Loss and Accuracy

Training and validation loss and accuracy were plotted. The curves showed that:

- Training and validation accuracy increased steadily
- Loss decreased over time
- No severe overfitting occurred (curves remained close)

7.2 Overfitting Analysis

Overfitting was minimal due to:

- Data augmentation
- Dropout layer
- Early stopping

7.3 Final Performance

The model achieved a strong validation accuracy, demonstrating effective learning and generalization.

7.4 Early Stopping Effect

Training stopped early before 50 epochs when validation loss stopped improving, saving time and preventing overfitting.

8. Prediction Demonstration

Five test images were selected and evaluated. For each image:

- Predicted label
- Ground truth label
- Confidence percentage

The model correctly classified most images with high confidence, demonstrating good performance on unseen data.

9. Discussion & Reflection

Advantages of MobileNet

MobileNet provides fast training and good accuracy by using pretrained weights and efficient architecture, making it suitable for limited datasets.

Role of Data Augmentation

Data augmentation improved model generalization by introducing variability in training data, reducing overfitting and improving robustness.

Possible Improvements

With more time and resources:

- Fine-tune deeper layers of MobileNet
- Use larger datasets
- Try advanced models like EfficientNet
- Perform hyperparameter tuning

10. Conclusion

This assignment successfully demonstrated the use of transfer learning with MobileNet for wildlife image classification. The model achieved good performance with efficient computation, showing the effectiveness of pretrained models and data augmentation in deep learning tasks.

List of Libraries used:

- tensorflow
- matplotlib
- numpy
- scikit-learn

Model Summary

```
model.summary()
```

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
sequential (Sequential)	(None, 224, 224, 3)	0
true_divide (TrueDivide)	(None, 224, 224, 3)	0
subtract (Subtract)	(None, 224, 224, 3)	0
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 128)	163,968
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

```
Total params: 2,423,242 (9.24 MB)
Trainable params: 165,258 (645.54 KB)
Non-trainable params: 2,257,984 (8.61 MB)
```

The 5 prediction images (as saved in the figure):

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 1s/step
```

Predicted: pecora (100.0%)
Ground Truth: pecora

```
Image 1: Predicted = pecora (100.0%), Ground Truth = pecora
✅
```

```
--------------------------------------------------
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step
```

Predicted: ragno (100.0%)
Ground Truth: ragno

```
Image 2: Predicted = ragno (100.0%), Ground Truth = ragno
✅
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step
```

Predicted: gallina (91.0%)
Ground Truth: gallina

```
Image 3: Predicted = gallina (91.0%), Ground Truth = gallina
✅
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 57ms/step
```

Predicted: cane (81.1%)
Ground Truth: cane

```
Image 4: Predicted = cane (81.1%), Ground Truth = cane
✅
```

```
--------------------------------------------------
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 51ms/step
```

Predicted: pecora (99.5%)
Ground Truth: pecora

```
Image 5: Predicted = pecora (99.5%), Ground Truth = pecora
✅
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```